Lecture 11

When is an error correctable?

Objectives. *Lecture 10 protected a qubit against every one-qubit error, with a recovery built for Shor's code. This lecture asks:* which errors can a code correct? *The answer is one condition on the code and the errors, Knill and Laflamme's: if* $\Pi_C E_i^\dagger E_j \Pi_C = \alpha_{ij}\Pi_C$ *for numbers* α_{ij}*, a recovery exists (Theorem 11.2, the main result; the converse is cited). A recovery that corrects a set of errors undoes every noise whose Kraus operators are combinations of them (Theorem 11.5), Lecture 10's discrete collapse in general.*

By the end of this lecture you will be able to:

- State the Knill–Laflamme condition, prove that it gives a recovery that corrects the errors, and state its converse with its source.
- Prove that a recovery that corrects a set of errors undoes every noise whose Kraus operators are linear combinations of them.
- Compute the Knill–Laflamme matrix of the bit-flip code and of Shor's code for their errors, and say when a code is degenerate.

11.1 Recovery as a channel

In Lecture 10 a recovery measured a syndrome and applied a correction. Averaged over the outcome it is $\mathcal{R}(\tau) = \sum_s C_s \Pi_s \tau \Pi_s C_s^\dagger$: a map with the Kraus operators $R_s = C_s\Pi_s$, and $\sum_s R_s^\dagger R_s = \sum_s \Pi_s = \mathbb{1}$. The question needs recoveries of every kind.

Definition 11.1 (Recovery, correction)**.** A *code* is a subspace $C \neq 0$ of $\mathbb{C}^{2^n}$, for a register of n qubits, with the projector Π_C; *errors* are any matrices $E_1, \dots, E_m$ on the register. A *recovery* is a map $\mathcal{R}(\tau) = \sum_l R_l \tau R_l^\dagger$ with $\sum_l R_l^\dagger R_l = \mathbb{1}$. It *corrects* $E_1, \dots, E_m$ on C if there are numbers λ_{li} with

$$R_l E_i \Pi_C = \lambda_{li} \Pi_C$$

for every l and i.

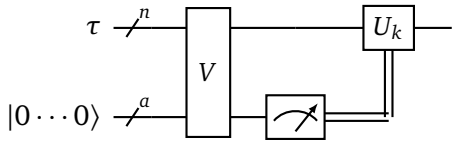
Then every density operator ρ on C, that is with $\Pi_C \rho \Pi_C = \rho$, has $\mathcal{R}(E_i \rho E_i^\dagger) = \sum_l |\lambda_{li}|^2 \rho$: the error is undone, up to the weight it had. Lecture 10's recoveries correct $\mathbb{1}$, X_1, X_2, X_3 on the bit-flip code and the 28 errors of Shor's code (the proofs of Theorem 10.4 and Theorem 10.6). The errors need not form a noise: $\mathbb{1}$, X_1, X_2, X_3 have $\sum_i E_i^\dagger E_i = 4\mathbb{1}$. The condition $\sum_l R_l^\dagger R_l = \mathbb{1}$ cannot be dropped: the map $\tau \mapsto 0$ satisfies the display with every $\lambda_{li} = 0$, and undoes nothing.

The definition already gives a necessary condition. If $\mathcal{R}$ corrects the errors, then

$$\begin{aligned} \Pi_C E_i^\dagger E_j \Pi_C &= \textstyle\sum_l (R_l E_i \Pi_C)^\dagger R_l E_j \Pi_C \\ &= \textstyle\sum_l \overline{\lambda_{li}} \lambda_{lj} \Pi_C, \end{aligned}$$

inserting $\sum_l R_l^\dagger R_l = \mathbb{1}$: a multiple of Π_C for every i and j. §11.2 proves that this is also enough.

A recovery of the form $R_k = U_k \Pi_k$, with a projective measurement $\{\Pi_k\}$ and gates U_k, is a procedure: measure, then correct. As in Lemma 9.3, the measurement is a gate V that writes k onto fresh wires, $\psi \otimes |0\cdots0\rangle \mapsto \sum_k \Pi_k\psi \otimes |k\rangle$, and a reading:



By Theorem 4.5 the reading can wait until after the corrections, controlled by the record: the recovery is then one gate on the register and fresh wires, which are discarded, as the environment was in Definition 10.1 (Nielsen and Chuang, Box 10.1, p. 439).

11.2 KL sufficiency

Theorem 11.2 (Knill–Laflamme). *Let C be a code and $E_1, \dots, E_m$ errors. If there are numbers α_{ij} with*

$$\Pi_C E_i^\dagger E_j \Pi_C = \alpha_{ij} \Pi_C$$

for all i and j, then some recovery corrects $E_1, \dots, E_m$ on C.

The condition says that the errors move every code vector alike: $\langle E_i\varphi | E_j\psi\rangle = \alpha_{ij} \langle\varphi|\psi\rangle$ for all φ, ψ in C. The proof turns the errors into combinations that take C to mutually orthogonal copies of itself, and undoes each copy.

Proof. *The matrix.* For a unit vector ψ in C, $\alpha_{ij} = \langle\psi| \Pi_C E_i^\dagger E_j \Pi_C |\psi\rangle = \langle E_i\psi | E_j\psi\rangle$. So $\alpha_{ji} = \overline{\alpha_{ij}}$, and $\sum_{i,j} \overline{c_i} c_j \alpha_{ij} = \left\| \sum_j c_j E_j \psi \right\|^2 \geq 0$ for all numbers c_j: α is self-adjoint with eigenvalues $d_k \geq 0$. By the spectral theorem, $\alpha = u d u^\dagger$ with u unitary and $d = \operatorname{diag}(d_1, \dots, d_m)$.

Orthogonal errors. Put $F_k = \sum_i u_{ik} E_i$. Then

$$\begin{aligned} \Pi_C F_k^\dagger F_l \Pi_C &= \textstyle\sum_{i,j} \overline{u_{ik}} u_{jl}\, \alpha_{ij} \Pi_C \\ &= d_k \delta_{kl} \Pi_C. \end{aligned}$$

So F_k multiplies the length of every code vector by $\sqrt{d_k}$, different F_k take C to mutually orthogonal subspaces, and $F_k \Pi_C = 0$ if $d_k = 0$. As u is unitary, $E_i = \sum_k \overline{u_{ik}} F_k$.

The recovery. For each k with $d_k > 0$ let $W_k = F_k \Pi_C / \sqrt{d_k}$. By the display, $W_k^\dagger W_l = \delta_{kl} \Pi_C$ for such k and l. So $\Pi_k = W_k W_k^\dagger$ is self-adjoint with $\Pi_k^2 = W_k \Pi_C W_k^\dagger = \Pi_k$, as $W_k \Pi_C = W_k$: the projector onto $F_k C$. As $\Pi_k \Pi_l = 0$ for $k \neq l$, $\Pi_\perp = \mathbb{1} - \sum_k \Pi_k$ is a projector too. Let the recovery have the Kraus operators $W_k^\dagger$, for every k with $d_k > 0$, and $\Pi_\perp$:

$$\begin{aligned} \textstyle\sum_k W_k W_k^\dagger + \Pi_\perp^\dagger \Pi_\perp &= \textstyle\sum_k \Pi_k + \Pi_\perp \\ &= \mathbb{1}. \end{aligned}$$

It corrects. For $d_k > 0$, $W_k^\dagger F_l \Pi_C = \Pi_C F_k^\dagger F_l \Pi_C / \sqrt{d_k} = \sqrt{d_k}\, \delta_{kl} \Pi_C$, so

$$W_k^\dagger E_i \Pi_C = \overline{u_{ik}} \sqrt{d_k}\, \Pi_C.$$

And $\Pi_\perp E_i \Pi_C = 0$, since each $F_l \Pi_C$ is 0 or $\sqrt{d_l}\, W_l = \sqrt{d_l}\, \Pi_l W_l$, which $\Pi_\perp$ annihilates. ∎

The recovery is a procedure: measure $\{\Pi_k, \Pi_\perp\}$; on the outcome k, apply a gate U_k that agrees with $W_k^\dagger$ on the range of Π_k, which $W_k^\dagger$ maps isometrically onto C (complete orthonormal bases, as in Lemma 9.3); on $\Pi_\perp$, do nothing. Then $U_k \Pi_k = W_k^\dagger \Pi_k = W_k^\dagger$.

Example 11.3 (Shor's code, $\mathbb{1}$, Z_1, Z_2)**.** The errors Z_1 and Z_2 take the code to subspaces orthogonal to it, and $Z_1 Z_2$ fixes every code vector (§10.4), so $\Pi_C Z_1 Z_2 \Pi_C = \Pi_C$ and

$$\alpha = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 1 \\ 0 & 1 & 1 \end{pmatrix},$$

with the eigenvalues 1, 2 and 0 and the eigenvectors $(1,0,0)$, $(0,1,1)/\sqrt{2}$ and $(0,1,-1)/\sqrt{2}$. The construction gives three cases:

1. For $d = 1$,
$$F = \mathbb{1},$$
$$W = \Pi_C.$$
2. For $d = 2$,
$$F = \frac{Z_1 + Z_2}{\sqrt{2}},$$
$$F\Pi_C = \sqrt{2}\, Z_1 \Pi_C,$$
$$W = Z_1 \Pi_C,$$
since $Z_2 \Pi_C = Z_1 \Pi_C$.
3. For $d = 0$,
$$F = \frac{Z_1 - Z_2}{\sqrt{2}},$$
$$F\Pi_C = 0.$$
Drop this direction before dividing by $\sqrt{d}$.

The recovery has the Kraus operators

$$\Pi_C,$$
$$\Pi_C Z_1,$$
$$\Pi_\perp.$$

Here $\Pi_\perp$ projects onto the 508 dimensions orthogonal to C and $Z_1 C$. The recovery asks whether the state is in C, in $Z_1 C$ or elsewhere, and applies Z_1 in the second case. It never tells Z_1 from Z_2, and need not.

11.3 The converse

> **Theorem 11.4** (The converse · *cited*)**.** *If a recovery $\mathcal{R}$ has $\mathcal{R}(\mathcal{N}(\rho)) = \rho$ for every density operator ρ on C, for a noise $\mathcal{N}(\tau) = \sum_i E_i \tau E_i^\dagger$ with $\sum_i E_i^\dagger E_i = \mathbb{1}$, it corrects $E_1, \ldots, E_m$, and so they meet the condition of Theorem 11.2.*

It rests on a theorem on Kraus operators not proved here (Nielsen and Chuang, Theorem 8.2 and pp. 437–438; Knill and Laflamme, p. 14).

11.4 Discretisation

Theorem 11.5 (Discretisation). *Let a recovery $\mathcal{R}$ correct $E_1, \dots, E_m$ on C. Then it corrects every linear combination of them, and every noise $\mathcal{N}(\tau) = \sum_k K_k \tau K_k^\dagger$, $\sum_k K_k^\dagger K_k = \mathbb{1}$, whose K_k are linear combinations of the E_i has $\mathcal{R}(\mathcal{N}(\rho)) = \rho$ for every density operator ρ on C.*

Proof. For $K = \sum_i c_i E_i$, $R_l K \Pi_C = \sum_i c_i \lambda_{li} \Pi_C$. So for the noise $R_l K_k \Pi_C = \mu_{lk} \Pi_C$ with numbers μ_{lk}, and as $\rho = \Pi_C \rho \Pi_C$,

$$\mathcal{R}(\mathcal{N}(\rho)) = \textstyle\sum_{l,k} |\mu_{lk}|^2 \, \rho.$$

Both maps keep the trace, so the sum is $\operatorname{Tr} \rho = 1$, as in the proof of Theorem 10.6. ∎

With Theorem 11.2, a code that meets the condition for finitely many errors is protected against every noise whose Kraus operators lie in their span: a continuum. For Shor's code and its 28 errors this is Theorem 10.6(ii), since every 2×2 matrix is a combination of $\mathbb{1}$, X, Y, Z. Nielsen and Chuang's Theorem 10.2 (p. 438) states it for the recovery their proof constructs; here it holds for every recovery that corrects the errors. A recovery known only to undo one noise corrects its Kraus operators by Theorem 11.4, and then every combination of them.

11.5 R's matrix

For the bit-flip code and the errors $\mathbb{1}$, X_1, X_2, X_3: if $i \neq j$, $E_i^\dagger E_j$ flips one or two bits of 000 and of 111, taking C to a subspace orthogonal to it, so $\Pi_C E_i^\dagger E_j \Pi_C = 0$; and $E_i^\dagger E_i = \mathbb{1}$. Thus the Knill–Laflamme matrix is

$$\alpha = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}.$$

Every $d_k = 1$. With $u = \mathbb{1}$, the orthogonal errors are $F_k = E_k$, and $\Pi_k = E_k \Pi_C E_k$ are the four syndrome projectors of Lecture 10. They add up to $\mathbb{1}$, so

$$\Pi_\perp = 0.$$

The recovery's Kraus operators are

$$\begin{aligned} W_k^\dagger &= \Pi_C E_k \\ &= E_k \Pi_{s(E_k)}. \end{aligned}$$

These are Lecture 10's $C_s \Pi_s$: the theorem rebuilds that recovery.

The set must be named. $\mathbb{1}$, X_1, X_2, X_3 is a set of errors, not a noise. Nielsen and Chuang's Exercise 10.7 (p. 438) weights them by $\sqrt{(1-p)^3}$ and $\sqrt{p(1-p)^2}$, which gives $\alpha = \operatorname{diag}((1-p)^3, p(1-p)^2, p(1-p)^2, p(1-p)^2)$ and $\sum_i E_i^\dagger E_i = (1 - 3p^2 + 2p^3)\mathbb{1}$: the part of independent flips with at most one flip. Weights do not change whether the condition holds. For $0 < p < 1$ the whole noise of independent flips fails it: its Kraus operators include multiples of $X_1 X_2$ and of X_3, and

$$\begin{aligned} \Pi_C (X_1 X_2)^\dagger X_3 \Pi_C = {} & |000\rangle \langle 111| \\ & + |111\rangle \langle 000|, \end{aligned}$$

not a multiple of Π_C. By Theorem 11.4 no recovery undoes it: the failure $q > 0$ of Theorem 10.4 is not a weakness of Lecture 10's recovery.

11.6 The degenerate guard

Shor's code meets the condition for its 28 errors, as its recovery corrects them (§11.1). Two errors with different syndromes have $\alpha_{ij} = 0$, as they take C into orthogonal syndrome spaces. Not so Z_1 and Z_2: $Z_1 Z_2$ fixes every code vector (§10.4), so $\Pi_C Z_1^\dagger Z_2 \Pi_C = \Pi_C$, and for these two errors

$$\alpha = \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix},$$

which is singular: $(Z_1 - Z_2)\Pi_C = 0$. A code is *degenerate* for a set of errors if α is singular (Gottesman, p. 14): some combination of the errors annihilates the code, here because two errors act alike on it. Example 11.3 shows the construction dropping that combination.

For the 28 errors, α is 1 off the diagonal exactly for Z_i and Z_j in one block; it has rank 22, and six zero eigenvalues. The construction of Theorem 11.2 drops those six directions, as $F_k \Pi_C = 0$ there, and keeps 22 copies of C, as many as Lecture 10 has syndromes. Degeneracy costs nothing: the recovery need not tell Z_1 from Z_2, since they act alike. A classical code, by contrast, tells errors on different bits apart (Nielsen and Chuang, p. 444).

For subsets of these 28 errors, α is singular exactly when it is not diagonal: an entry 1 off the diagonal means that two errors act alike, and their difference annihilates C. In general neither implies the other: on the bit-flip code $\{X_1, X_1 + X_2\}$ has $\alpha = \left(\begin{smallmatrix} 1 & 1 \\ 1 & 2 \end{smallmatrix}\right)$, not diagonal and invertible, and on Shor's code $\{\mathbb{1}, Z_1 - Z_2\}$ has $\alpha = \mathrm{diag}(1, 0)$, diagonal and singular. Lecture 12 lets two errors that act alike share a syndrome.

Summary.

- *Recovery:* a map with Kraus operators R_l, $\sum_l R_l^\dagger R_l = \mathbb{1}$; it corrects $E_1, \ldots, E_m$ on C if $R_l E_i \Pi_C = \lambda_{li} \Pi_C$ (Definition 11.1).
- *The main result:* if $\Pi_C E_i^\dagger E_j \Pi_C = \alpha_{ij} \Pi_C$, some recovery corrects the errors (Theorem 11.2); a recovery that undoes a noise forces the condition on its Kraus operators (Theorem 11.4, cited).
- *Discretisation:* a recovery that corrects a set of errors undoes every noise whose Kraus operators are combinations of them (Theorem 11.5).
- *R:* $\alpha = \mathbb{1}$ for $\mathbb{1}, X_1, X_2, X_3$, and the construction is Lecture 10's recovery; independent flips, $0 < p < 1$, are not correctable.
- *Degenerate:* for Z_1, Z_2 on Shor's code α is singular; the recovery corrects both.
- *Next:* Lecture 12, stabilizer codes.

References

Nielsen and Chuang, pp. 435–441 and 444. E. Knill and R. Laflamme, *A theory of quantum error-correcting codes*, arXiv:quant-ph/9604034 (1996), pp. 12–15. D. Gottesman, *Stabilizer codes and quantum error correction*, arXiv:quant-ph/9705052 (1997), pp. 13–15.